

Date : 17/06/26

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Week-2 Module

1. Fraud Transaction Detection (Linear Search)

Problem: A bank stores transaction IDs in order they occurs. Since the transaction are not sorted, the security team needs to check if a suspicious transaction ID exists.

Code:

```
import java.util.Scanner;
public class LinearSearch {
    public static void main (String[] args) {
        Scanner s = new Scanner (System.in);
        int n = s.nextInt();
        int [] arr = new int[n];
        for (int i=0; i<n; i++) {
            arr[i] = s.nextInt();
        }
        int sus = s.nextInt();
        boolean found = false;
        for (int i=0; i<n; i++) {
            if (arr[i] == sus) {
                found = true;
                break;
            }
        }
    }
}
```

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```
if (found) {
```

```
    System.out.println("Fraud Transaction  
                        ID found");
```

```
} else {
```

```
    System.out.println("Transaction not found");
```

```
}
```

```
s.close();
```

```
}
```

```
}
```



```
1 import java.util.Scanner;
2
3 public class LinearSearch {
4     public static void main(String[] args) {
5
6         Scanner s = new Scanner(System.in);
7
8         int n = s.nextInt();
9         int[] arr = new int[n];
10
11         for(int i = 0; i < n; i++) {
12             arr[i] = s.nextInt();
13         }
14
15         int sus = s.nextInt();
16
17         boolean found = false;
18
19         for(int i = 0; i < n; i++) {
20             if(arr[i] == sus) {
21                 found = true;
22                 break;
23             }
24         }
25
26         if(found) {
27             System.out.println("Fraud Transaction Found");
28         } else {
29             System.out.println("Transaction Not Found");
30         }
31
32         s.close();
33     }
34 }
```

```
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
```

=== Code Execution Successful ===

2. Server Log Search (Binary Search)

Problem: A cloud company stores server log IDs in sorted order. Engineers need to quickly verify if a log ID exists.

Code:

```
import java.util.Scanner;
public class Binary Search {
    public static void main (String[] args) {
        Scanner s = new Scanner (System.in);
        int n = s.nextInt();
        int [] arr = new int [n];
        for (int i=0; i<n; i++) {
            arr[i] = s.nextInt();
        }
        int target = s.nextInt();
        int start = 0;
        int end = n - 1;
        boolean found = false;
```



```
while (start <= end) {  
    int mid = start + (end - start) / 2;  
    if (arr[mid] == target) {  
        found = true;  
        break;  
    }  
    else if (arr[mid] < target) {  
        start = mid + 1;  
    }  
    else {  
        end = mid - 1;  
    }  
}  
if (found) {  
    System.out.println("Log found");  
} else {  
    System.out.println("Log Not found");  
}  
s.close();
```

```
1 import java.util.Scanner;
2
3 public class BinarySearch {
4     public static void main(String[] args) {
5
6         Scanner s = new Scanner(System.in);
7         int n = s.nextInt();
8         int[] arr = new int[n];
9
10        for(int i = 0; i < n; i++) {
11            arr[i] = s.nextInt();
12        }
13
14        int target = s.nextInt();
15        int start = 0;
16        int end = n - 1;
17        boolean found = false;
18
19        while(start <= end) {
20            int mid = start + (end - start) / 2;
21            if(arr[mid] == target) {
22                found = true;
23                break;
24            }
25            else if(arr[mid] < target) {
26                start = mid + 1;
27            }
28            else {
29                end = mid - 1;
30            }
31        }
32        if(found) {
33            System.out.println("Log Found");
34        } else {
35            System.out.println("Log Not Found");
36        }
37        s.close();
38    }
39 }
```

```
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found
```

=== Code Execution Successful ===

3. Sort Employee Salaries (Bubble Sort)

Problem: A company wants to sort employee salaries in ascending order for payroll analysis.

Code:

```
import java.util.Scanner;
public class BubbleSort {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        int n = s.nextInt();
        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = s.nextInt();
        }
        for (int i = 0; i < n-1; i++) {
            for (int j = 0; j < n-i-1; j++) {
                if (arr[j] > arr[j+1]) {
                    int temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                }
            }
        }
        for (int n : arr) {
            System.out.print(n + " ");
        }
        s.close();
    }
}
```

```
1 import java.util.Scanner;
2
3 public class BubbleSort {
4     public static void main(String[] args) {
5
6         Scanner s = new Scanner(System.in);
7
8         int n = s.nextInt();
9         int[] arr = new int[n];
10
11         for(int i = 0; i < n; i++) {
12             arr[i] = s.nextInt();
13         }
14
15         for(int i = 0; i < n - 1; i++) {
16
17             for(int j = 0; j < n - i - 1; j++) {
18
19                 if(arr[j] > arr[j + 1]) {
20
21                     int temp = arr[j];
22                     arr[j] = arr[j + 1];
23                     arr[j + 1] = temp;
24                 }
25             }
26         }
27
28         for(int num : arr) {
29             System.out.print(num + " ");
30         }
31
32         s.close();
33     }
34 }
```

```
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
=== Code Execution Successful ===
```